

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

Anonymous authors
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ABSTRACT

Although diffusion probabilistic models (DPMs) have attained conspicuous advances across an expanding array of generative tasks, encompassing image and video synthesis alike, they persistently emit hallucinated samples (*hallucinations*)—outputs that contravene real-world regularities, as exemplified by the synthesis of an implausible duplicate cup suspended beside another. Notwithstanding the prevalence of such artifacts, the conspicuous absence of tractable methodologies for their systematic quantification simultaneously impedes principled progress toward remediation and obscures the pathways through which next-generation generative models might be endowed with factual constraints. The present work seeks to close this gap by concentrating on a circumscribed variety of hallucination, which we designate **counting hallucination**: the generation of an erroneous number of instances or structured objects—for instance, a hand rendered with six fingers—notwithstanding the demonstrable absence of any such configuration from the training distribution. To operationalize this notion, we assemble a dataset suite, **CountHalluSet**, furnished with rigorously specified counting criteria and comprising ToyShape, SimObject, and RealHand. Upon this foundation we erect a standardized evaluation protocol for quantifying counting hallucinations, through which we systematically interrogate how the diverse sampling conditions of DPMs—solver type, ODE solver order, the number of sampling steps, and the initial noise—modulate the incidence of counting hallucination. We further characterize the correlation between these hallucinations and prevailing evaluation metrics, notably the Fréchet Inception Distance (FID), thereby disclosing that widely adopted perceptual metrics fail to register counting hallucinations with any consistency.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), conventionally abbreviated as diffusion models, have accrued considerable success across the landscape of generative modeling, yielding high-fidelity samples in a heterogeneous collection of tasks that spans text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Yet, for all these advances, diffusion models remain conspicuously deficient in their capacity to faithfully internalize the factual regularities of the training data, and they routinely emit outputs that transgress consistency with real-world knowledge. A paradigmatic instance of this failure is the generation of objects endowed with an incorrect number of spatially intricate constituents—supernumerary human fingers (Narasimhaswamy et al., 2024), say, or additional animal limbs (Borji, 2023).

A burgeoning body of recent scholarship has trained its attention on the identification and analysis of the failure modes that recur in text-guided diffusion models. Borji (2023), for one, offered a systematic catalogue of the qualitative shortcomings that beset text-to-image models. In a complementary vein, Liu et al. (2024) advanced an adversarial search procedure designed to expose undesirable behaviors and failure cases in text-to-image generation, whereas Wu et al. (2023) adopted an iterative methodology for the discovery and subsequent mitigation of spurious correlations.

More recently still, *hallucination*—a failure mode peculiar to diffusion models—has begun to command the sustained attention of the research community (Aithal et al., 2024). In contradistinction

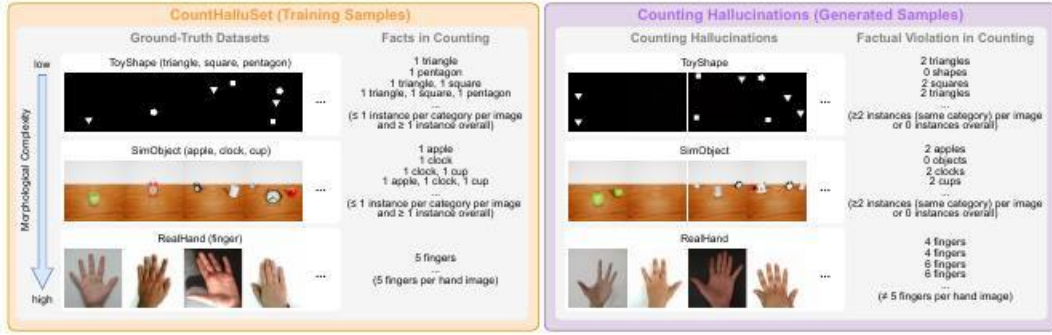


Figure 1: Illustrative juxtaposition of the proposed **CountHalluSet** employed for training (left) against the **counting hallucinations** the model subsequently generates (right). Each constituent dataset comprises distinct objects governed by its own counting criteria: in ToyShape and SimObject, every image contains at most one instance per category and at least one instance in aggregate, whereas in RealHand every image contains precisely five fingers. Counting hallucinations denote generated images that contravene the dataset’s counting rules—two apples within a single image, say, or an image devoid of objects yet furnished with an otherwise normal background—even though no such configuration is ever exhibited in the SimObject dataset.

to the text-prompt-based failure modes canvassed above, whose severity is amenable to assessment through the alignment between input conditions and outputs (as gauged, for example, by the CLIP score (Hessel et al., 2021) or the VQA score (Huang et al., 2025; Zarei et al., 2024)), and in contradistinction likewise to the more overt failures of severe distortion, blur, and artifacting, hallucinated samples (*hallucinations*) frequently present as visually plausible and arise independently of the conditioning signal; they surface instead as subtle transgressions of factual consistency with respect to the training data—incorrect object counts, geometries that appear plausible yet remain physically unrealizable, or comparable breaches of physical law. Such fine-grained distinctions between valid samples and their hallucinated counterparts constitute a formidable challenge for contemporary AI systems, which frequently prove incapable of detecting them with any reliability. Indeed, even state-of-the-art large vision-language models of the caliber of GPT-4o continue to founder on tasks as elementary as enumerating the fingers in an unambiguous hand image. It follows that the establishment of a standardized and rigorously delineated protocol for the reliable quantification of hallucinated samples is not merely desirable but essential, inasmuch as it furnishes the requisite foundation for charting the design of next-generation generative models governed by strict factual consistency—a competence indispensable to their prospective role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Of the manifold varieties of hallucinated element—counts, colors, spatial relations, geometry, and physical realism among them—counting errors commend themselves as peculiarly amenable to systematic study, for they are at once readily identifiable and unambiguously quantifiable. This consideration motivates us to inaugurate our inquiry by delimiting a specific form of hallucination that we term **counting hallucination**. *Within this setting, the number of objects present in the training data is explicitly predetermined; a generated image is accordingly adjudged hallucinated precisely when its object counts depart from these prescribed values.* An image depicting a hand with six fingers, for instance, is classified as a counting hallucination, inasmuch as no such configuration ever materializes in the RealHand dataset, as the last row of Fig. 1 attests. To render the study of such phenomena tractable, we construct a dataset suite, CountHalluSet, comprising three datasets that together traverse a spectrum of morphological complexity in the countable objects: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 exhibits representative examples drawn from these datasets alongside their corresponding counting hallucinations.

Within our quantification protocol for counting hallucinations, the pivotal step resides in the accurate identification and enumeration of objects, a task we discharge by means of counting models tailored to each dataset. Marshalling these datasets and counting models in concert, we examine how the various denoising conditions internal to diffusion models—the number of sampling steps, the initial noise, the order of the ODE solver, and the solver type—bear upon counting hallucination rates. In addition, we undertake a correlation analysis relating the counting hallucination rate to perceptual

metrics, among them the Fréchet Inception Distance (FID) [Heusel et al. \(2017\)](#) and MUSIQ [Ke et al. \(2021\)](#).

Our principal findings and contributions may be enumerated as follows:

- The numerical strategies customarily deployed in diffusion models—augmenting the number of sampling steps from 25 to 100, for instance—frequently fail to attenuate counting hallucinations in the generation of human hands, notwithstanding the efficacy these same techniques exhibit on simpler synthetic datasets.
- Only a weak correlation obtains between the counting hallucination rate and perceptual metrics, whereas a markedly strong correlation emerges between the non-counting failure rate and those same metrics.
- Prompted by one of these findings, we propose a novel and auspicious training paradigm—the joint-diffusion model—which substantially curtails both counting-based hallucinations and non-counting failures on the RealHand dataset.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Galvanized by the mounting scrutiny that hallucination has attracted in the setting of large language models (LLMs) ([Kadavath et al., 2022](#); [Ji et al., 2023](#); [Huang et al., 2023](#)), the study of hallucination in unconditional diffusion models has likewise begun to elicit escalating interest. [Aithal et al. \(2024\)](#), in particular, advanced a qualitative hypothesis to the effect that hallucinated samples may originate in the interpolation of data between distinct modes. Illuminating as this conjecture regarding potential causes may be, a systematic quantitative analysis of hallucination in diffusion models remains conspicuously wanting.

This lacuna motivates our endeavor to facilitate the accurate identification and quantification of hallucinations in diffusion-generated outputs. An investigation of this kind promises both to deepen our understanding of the undesirable behaviors these models exhibit and to elucidate how the magnitude of the denoising error—as modulated, for example, by variation in the number of sampling steps or in the order of the ODE solver—governs the severity of the resulting hallucinations.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score ([Huang et al., 2025](#); [Zarei et al., 2024](#)) and the CLIP score ([Hessel et al., 2021](#)) rank among the most widely adopted metrics for appraising the performance of text-guided image generation. Evaluation schemes of this description, however, predicated as they are upon language-based models, are intrinsically circumscribed when enlisted to assess hallucinated images. One strand of evidence to this effect is the statistical lower bound on the propensity of pretrained language models to hallucinate certain classes of fact, a bound that in turn caps the detection accuracy attainable by any language-model-based approach ([Kalai & Vempala, 2024](#)). A second indication resides in the persistent performance gap that separates human annotators from LLMs in the classification of hallucinated textual content ([Lin et al., 2022](#); [Li et al., 2023](#)).

Beyond these, a number of extensively used quantitative evaluation metrics—FID ([Heusel et al., 2017](#)), IS ([Salimans et al., 2016](#)), Precision & Recall ([Sajjadi et al., 2018](#)), and MUSIQ ([Ke et al., 2021](#))—are routinely invoked in image generation tasks, each deriving from the feature embeddings extracted by a pretrained Inception-v3 model ([Szegedy et al., 2016](#)). Whether such metrics are in fact capable of capturing the severity of hallucinations in generated datasets nonetheless *remains an open question*, one that the present work sets out to address from the vantage point of counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) posit a *forward process* (equivalently, a *diffusion process*) that progressively corrupts the data through the injection of Gaussian noise over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample $\mathbf{x}_t$, conditioned upon the original data $\mathbf{x}_0$, is given by:

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_t\}_{t=1}^T$ constitutes a predefined, t -dependent noise schedule, conventionally instantiated as either a linear (Ho et al., 2020) or a cosine (Nichol & Dhariwal, 2021) schedule. The factor $\bar{\alpha}_t$ regulates the signal-to-noise ratio at step t , ranging from $\bar{\alpha}_0 = 1$ (an absence of noise) to $\bar{\alpha}_T \approx 0$ (a state of near-pure noise). An equivalent reformulation of Eq. (1) reads:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

As T grows sufficiently large, the marginal distribution $q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ converges toward a standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In the DPMs of widest currency (Ho et al., 2020; Nichol & Dhariwal, 2021), T is customarily fixed at 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (equivalently, the *denoising process*) of DPMs is charged with iteratively reconstructing the observed data from the diffused data $\mathbf{x}_T$. To this end, DPMs endeavor to predict the noise $\boldsymbol{\epsilon}$ from the data $\mathbf{x}_t$ by means of a neural network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, designated the *noise prediction model*. Its parameter θ is optimized through minimization of the following objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\pi(t) \|\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) - \boldsymbol{\epsilon}\|_2^2], \quad (3)$$

where $\pi(t) > 0$ denotes a weighting function.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, samples from the original distribution $q_0(\mathbf{x}_0)$ may be procured by reversing the diffusion process over an identical number of steps, a procedure conventionally termed *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1 - \beta_t}} (\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

wherein $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ constitutes the sole unknown term, designated the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1 - \bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ by way of the noise prediction model, whence an equivalent formulation of Eq. (3) follows:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1 - \beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \beta_t}} \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. Whereas the ancestral sampling of Eq. (4) harbors a stochastic term (Kloeden et al., 1992), there exists an associated deterministic process amenable to formulation as an ordinary differential equation (ODE), otherwise known as the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau) \mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2} \beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2} \beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE under the true score function $h(\mathbf{x}(\tau), \tau)$ and originates from ideal diffused data $\mathbf{x}(T)$ drawn from the true diffused

distribution $q_T(\mathbf{x}(T))$. Fixing the discrete time steps as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with $\tau_N = 0$, and invoking the variation-of-constants formula for the exact solver over a single step yields:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k)\mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u)h(\mathbf{x}(u), u)du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v)dv}$ is the state transition operator associated with the linear part $\frac{d\mathbf{z}}{d\tau} = f(\tau)\mathbf{z}$. Here $\Delta t > 0$ denotes the step size, whose enlargement relative to ancestral sampling is precisely what permits ODE-based solvers to accelerate the sampling process; the total number of sampling steps is correspondingly $T/\Delta t$. Because the linear part $\mathcal{G}(t_2, t_1)$ admits a closed-form evaluation, the preponderance of accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate their numerical resources—higher solver orders and additional sampling steps alike—upon the faithful approximation of the nonlinear integral term involving h . Further theoretical analysis is deferred to Appendix A.1 through A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section expounds the methodology and datasets through which counting hallucinations are quantified. We begin by formalizing the counting criteria and the counting hallucinations they induce, and proceed thereafter to introduce the datasets constituting the CountHalluSet suite. For each dataset in turn, we delineate its counting criteria and the corresponding counting hallucinations, and set forth the standardized protocol governing their quantification.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

In this section we furnish a general definition of the counting criteria that govern a training dataset, and characterize counting hallucinations as departures from these predefined constraints.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the number of objects of category c occurring in $\mathbf{x}$, namely $N_c(\mathbf{x})$, is constrained to lie within a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$); that is, $N_c(\mathbf{x}) \in \mathcal{S}_c$. We stipulate, in addition, that any valid reference sample contain at least one object—formally, $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C designates the set of categories under consideration.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we say that $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ precisely when both of the following conditions are satisfied:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must attain a degree of visual quality sufficient for the background and any objects present to be clearly discernible—a prerequisite that serves to insulate counting hallucinations from other salient failure modes, such as severe degradation. Accordingly, we introduce a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, termed the *counting-ready indicator*, in which 1 signals that the image quality suffices for counting and 0 signals the contrary. In practice this indicator may be adjudicated through human annotation, through a classifier calibrated against such annotation, or through alternative automated criteria.
2. **Violation of counting facts.** A violation of counting facts is incurred whenever there exists a category c for which $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or whenever the image is altogether devoid of objects (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$), given that the dataset mandates at least one object per sample. In practice we enlist a counting model to predict object counts in generated images, thereby furnishing evidence of such violations.

A counting hallucination (CH) is thus adjudged present whenever the image is of quality sufficient for counting ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and the predicted counts depart from the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluset suite is composed of three datasets devised for the study of counting hallucinations: ToyShape, SimObject, and RealHand. Representative examples from each dataset, together with their attendant counting hallucinations, are displayed in Fig. A1 and A2. Supplementary statistical summaries are consigned to Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset comprises 30,000 images populated by three geometric shapes: *triangle*, *square*, and *pentagon*. All shapes are rendered in white, occupy an identical area of 120 pixels, and are arranged so as never to overlap within any single image. The counting criteria prescribe that each image contain at most one instance of each shape category yet at least one shape in aggregate, as depicted in the top-left of Fig. 1. Formally, $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$, and for each class $c \in C$ we impose $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. By the counting criteria stipulated above, any image produced by a model trained on ToyShape that harbors two or more shapes of a single category constitutes a counting hallucination—the image bearing two pentagons in the top-right region of Fig. 1 being a case in point. Empty images, moreover—those exhibiting only the black background unaccompanied by any shape—are likewise classified as counting hallucinations. Illustrative examples are provided in Appendix A.5.

Counting model used in ToyShape. We deploy a counting model to detect counting errors within images while dispensing with the counting-ready indicator, since the dataset’s simplicity renders severely degraded images a rarity. Concretely, we fine-tune a ResNet-50 (He et al., 2016) on upwards of 400k ToyShape samples, in which each category appears between zero and three times per image (i.e., $S_c = \{0, 1, 2, 3\}$), the intent being to encompass a broad range of plausible scenarios—counting-valid samples and counting hallucinations alike—and thereby to enable accurate quantification. The resulting model attains a counting accuracy in excess of 99.9% on generated data.

Quantification of counting hallucinations. Given a diffusion model trained on the ToyShape dataset, we generate a collection of samples equal in cardinality to the training set (30,000 images). We then invoke the counting model to predict the number of objects in each generated image, and finally designate as counting hallucinations those images that transgress the counting criteria, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset encompasses 30,000 rendered images of everyday objects, drawn from three object categories—*mug*, *apple*, and *clock*—each furnished with 10 distinct intra-class variations. All objects are disposed at random upon a wooden tabletop under fixed lighting conditions, with minimal mutual occlusion where any occurs at all, as demonstrated in the central-left portion of Fig. 1. The dataset is generated synthetically within Unreal Engine 5 (Epic Games, 2024), whose photorealistic rendering affords an accurate simulation of object geometry, material attributes, lighting, and reflection. The counting criteria for this dataset parallel those of ToyShape (see Sec. 4.2.1): formally, $C = \{\text{mug}, \text{apple}, \text{clock}\}$, and for each class $c \in C$ we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Owing to the close affinity of their counting criteria, the definition of counting hallucinations, the construction of the counting model, and the quantification protocol all proceed exactly as in the ToyShape dataset, as shown in Fig. 2.

4.2.3 REALHAND

Dataset description. To interrogate counting hallucinations in real-world settings, we construct the RealHand dataset, comprising 5,050 human hand images sourced from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². Concretely, we aggregate the raw images from these sources and

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

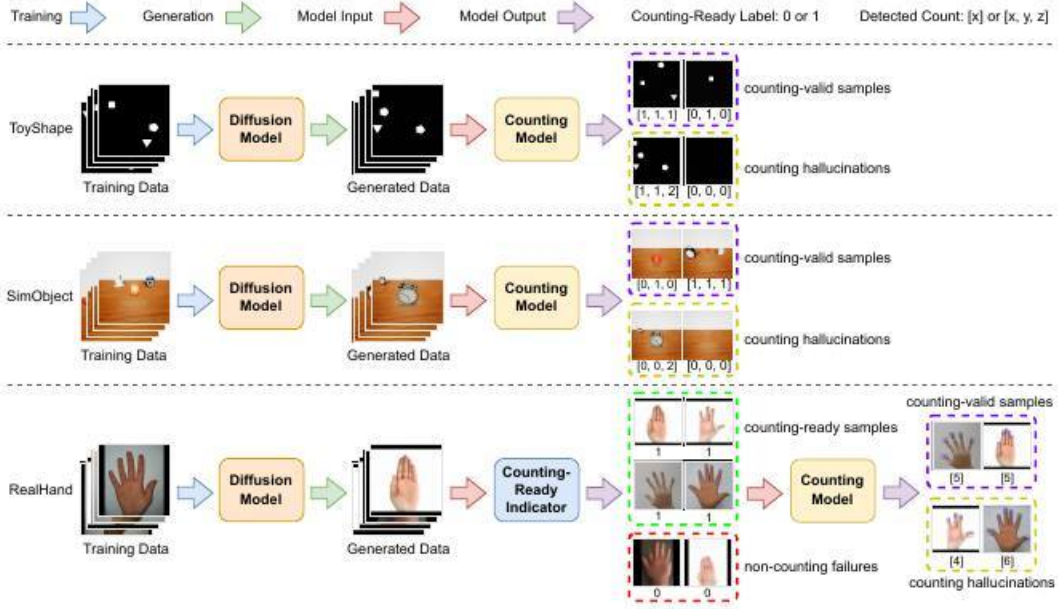


Figure 2: The evaluation procedure within the CountHalluSet suite. For ToyShape and SimObject, generated images are subjected directly to counting. For RealHand, by contrast, a counting-ready indicator—a binary classifier concordant with human judgment—is interposed so as to disentangle counting hallucinations from other, non-counting failures, such as severely deformed fingers. The count labels correspond to ToyShape: *triangle, square, pentagon*; SimObject: *mug, apple, clock*; and RealHand: *finger*.

expunge those marred by blur or artifacts. Each retained image depicts a human hand bearing five fingers, captured in both dorsal (back) and palmar (front) views under diverse lighting conditions and against varied backgrounds—white surfaces, blackboards, and walls among them—as shown in the lower-left of Fig. 1. The counting criteria prescribe that each image contain exactly five fingers; formally, $C = \{\text{finger}\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, whence $\sum_{c \in C} N_c(x) = 5$.

Counting hallucinations and non-counting failures. The generation of realistic hand images is inherently susceptible to unexpected failures that extend beyond counting hallucinations (such as supernumerary or missing fingers)—among them blurriness, severely deformed fingers, and visual artifacts. We therefore enlist a counting-ready indicator, in the form of a binary classifier, to identify such cases. Examples of counting hallucinations and of non-counting failure samples are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To segregate counting hallucinations from the failure modes just enumerated, we fine-tune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images, so that it may serve as a counting-ready indicator concordant with human judgment. For the counting task proper, we fine-tune a YOLO-12 (Tian et al., 2025) model on 2k fingertip-annotated generated images. As adjudged by the human evaluation of 500 randomly sampled images by three annotators, the indicator and the counting model attain accuracies in excess of 96% and 99%, respectively. Examples of YOLO detection results are provided in Appendix A.6.

Quantification of counting hallucinations. In contradistinction to the ToyShape and SimObject datasets, where the simplicity of the data and the concomitant rarity of severe failure cases (distorted or unrecognizable objects, for example) permit every generated image to be evaluated directly, the RealHand dataset necessitates a supplementary filtering step. Specifically, we first apply the counting-ready indicator to select hand images of sufficient visual quality, and only thereafter invoke the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For the ToyShape dataset we train a DDPM (Ho et al., 2020) from scratch, whereas for SimObject and RealHand we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on the CelebA-HQ dataset (Karras et al., 2018). The number of timesteps T is fixed at 1,000 under a standard linear noise schedule. Images are resized to 128×128 for the DDPMs and to 256×256 for the LDMs. Optimization proceeds via Adam (Kingma, 2015) at a learning rate of 0.0001, sustained for 150k, 300k, and 80k steps on the ToyShape, SimObject, and RealHand datasets, respectively, with an effective batch size of 256.

Inference of Diffusion Models. For ancestral sampling (i.e., DDPM), the number of sampling steps is set to 1,000. For the ODE-based solvers, we consider methods of both first and second order: DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)) as the first-order solver, and DPM-Solver-2 (Lu et al., 2022) as its second-order counterpart. Each solver is evaluated across three widely adopted denoising-step configurations—25, 50, and 100 steps. On the RealHand dataset, the counting model relies upon a YOLO detector configured with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless stated otherwise, the number of generated samples is matched to the number of training samples in each evaluation, and all reported results are averaged over three distinct random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 sets out the quantitative results for the counting hallucination rate (CHR), the non-counting failure rate (NCFR), and the total failure rate (TFR) under a range of denoising conditions—encompassing solver types, ODE solver orders, and initial-noise settings—across the three datasets ToyShape, SimObject, and RealHand. Several salient observations may be adduced.

(1) **Increasing the number of sampling steps attenuates counting hallucinations on synthetic datasets yet aggravates them on RealHand.** As Table 1 attests, under the commonly employed ODE solver settings (25, 50, and 100 steps), augmenting the number of sampling steps generally diminishes the counting hallucination rate on ToyShape and SimObject, but tends to inflate it on RealHand—the sole exception being 100 steps with DPM-Solver-2. This divergent behavior intimates that synthetic datasets, by virtue of their simpler and more structured distributions, profit from finer solver granularity, whereas real-world datasets such as RealHand possess distributions of greater complexity, in which additional steps may overfit local inconsistencies and thereby amplify hallucinations.

(2) **Recourse to higher-order ODE solvers effectively curtails non-counting failures yet exacerbates counting hallucinations on RealHand under the standard inference budget.** As Table 1 reveals, juxtaposing DPM-Solver-1 with DPM-Solver-2 on the RealHand dataset at the standard sampling budgets (25 and 50 steps, for instance) exposes a consistent trade-off: the higher-order solver, while substantially reducing non-counting failures, concurrently precipitates a higher counting hallucination rate. An analogous trend surfaces as the number of sampling steps is increased for both DPM-Solver-1 and DPM-Solver-2. In sum, the expenditure of additional computation—that is, the adoption of ostensibly “better” sampling conditions—consistently improves overall perceptual quality (as reflected in lower non-counting failure rates) but does nothing to enforce the factual correctness of finger counts.

(3) **DDPM substantially surpasses the ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across the full slate of comparisons, ancestral sampling (i.e., DDPM) uniformly attains the lowest CHR, NCFR, and TFR, which suggests that DDPM furnishes a practical lower bound for the failure rates of generative models. These findings imply that ancestral sampling constitutes an effective strategy for the minimization of hallucinations.

(4) **A more judicious initial-noise configuration can reduce both counting hallucinations and non-counting failures.** As Table 1 shows, the use of “diffused” noise (i.e., ground-truth initial noise) tends to yield lower CHR, NCFR, and TFR than does “normal” noise (i.e., standard Gaussian noise). This effect plausibly arises because diffused noise more faithfully aligns with the initial-noise distribution encountered during training, which in turn intimates that a more informed initialization can mitigate not only counting hallucinations but also failures of a non-counting character.

Table 1: Counting hallucination rates (CHR; the number of counting hallucinations normalized by the total number of generated samples) are reported under a variety of solver configurations and initial-noise settings across the three datasets ToyShape, SimObject, and RealHand. For RealHand we additionally report the non-counting failure rate (NCFR), which enumerates the failure cases apprehended by the counting-ready indicator. The total failure rate (TFR) is defined as the sum of CHR and NCFR. *Diffused* and *Normal* designate, respectively, the ground-truth initial noise and standard Gaussian noise.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

(5) **Greater morphological complexity in the countable objects heightens the propensity toward counting hallucination.** A comparison of CHR across ToyShape, SimObject, and RealHand indicates that diffusion models incur progressively higher CHR as the morphological complexity of the target objects mounts. This pattern suggests that the fine-grained detail and intricate spatial relations intrinsic to morphologically complex objects impose additional demands upon diffusion models seeking to preserve accurate object counts, thereby occasioning more frequent additions and omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Since counting hallucination is a species of factualness hallucination, it is natural to inquire: *Does a superior perceptual metric, such as a lower FID, necessarily entail fewer counting hallucinations by way of a diminished distributional gap between generated and training data?* The answer, we contend, is **not necessarily**.

To probe this relationship, we conduct a correlation analysis relating CHR to perceptual metrics, and likewise relating NCFR to the same metrics. We concentrate upon two extensively used perceptual measures: a distribution-based metric, FID (Heusel et al., 2017), and an instance-based metric, MUSIQ (Ke et al., 2021). For the estimation of correlation we adopt three complementary measures—the Pearson correlation (Lee Rodgers & Nicewander, 1988), which captures linear dependence; the Spearman rank correlation (Spearman, 1904), which captures monotonic dependence; and the Distance Correlation (Székely et al., 2007), which captures general nonlinear dependence. The evidence marshalled below substantiates our conclusion.

✚ **The correlations between CHR and perceptual metrics are contingent upon dataset and solver rather than intrinsic.** As the first and third rows of Table 2 indicate, the Pearson correlations between CHR and FID fluctuate substantially across datasets—positive for SimObject, negative for RealHand, and positive once more for PTI-Hand—while the corresponding Spearman correlations traverse a sequence from positive to negative to negative. The distance correlations likewise decline from 0.82 (SimObject) through 0.74 (RealHand) to 0.47 (PTI-Hand), a progression suggesting that more complex generative content attenuates the association between CHR and FID. The integration of DDPM results with those obtained from the ODE solvers, moreover, further weakens these correlations.

✚ **The correlations between NCFR and perceptual metrics are uniformly strong and positive.** In marked contrast to the tenuous correlations exhibited between CHR and perceptual metrics, those between NCFR and perceptual metrics prove strong and robust across every setting examined. For NCFR versus FID, for example, all correlation coefficients exceed 0.88 and are accompanied by exceedingly small p-values, betokening high statistical significance.

Table 2: Correlation of the counting hallucination rate (CHR), the non-counting failure rate (NCFR), and the total failure rate (TFR) with perceptual metrics, namely FID and MUSIQ. The results for PTI-Hand are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts so as to produce 5050 samples, thereby matching the size of the RealHand dataset. FID for PTI-Hand is computed between the RealHand images and the model-generated samples. Here p denotes the p-value gauging the statistical significance of the correlation r or ρ . The symbol $\dagger$ signifies that the DDPM results have been integrated with those from the various ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand	
	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01						
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—						
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—						
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—						
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—						

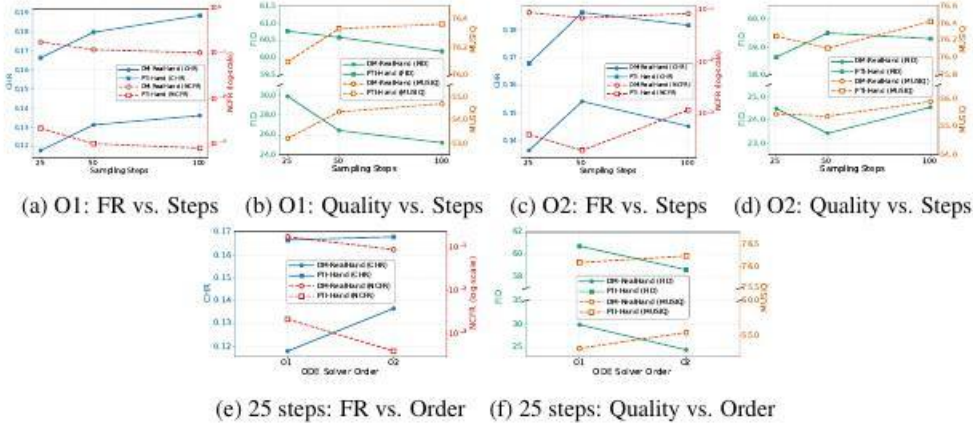


Figure 3: A comparison of trend consistency between DM-RealHand (ours) and PTI-Hand (pre-trained Stable-Diffusion-3.5-Medium). O1 and O2 denote the first- and second-order ODE solvers, respectively (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). The FR metrics comprise CHR and NCFR, whereas the quality metrics comprise FID and MUSIQ. Across variation in both the number of sampling steps and the solver order, the two models exhibit trends that are strikingly consistent with one another.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

In this section we appraise counting hallucinations within a large-scale pretrained text-to-image model. Specifically, we assess Stable-Diffusion-3.5-Medium (Esser et al., 2024) by means of the counting-ready indicator and counting model introduced in Sec. 4.2.3; further particulars are supplied in Appendix A.8. As Figure 3 demonstrates, the manner in which solver type and the number of sampling steps modulate counting hallucinations in large pretrained diffusion models aligns closely with the trends disclosed by our controlled RealHand experiments. Such consistency both corroborates the validity of the findings advanced in Sec. 5.2 and underscores the broader applicability of our evaluation protocol. The correlation results in Table 2 lend this conclusion additional support: here too, weak correlations between CHR and perceptual metrics coexist with strong correlations between NCFR and perceptual metrics. Taken together, these results attest to the robustness and generality of our findings, and demonstrate that the proposed evaluation protocol transfers reliably to pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The diminished counting hallucination rates observed for the simpler countable objects (see Observation 5 in Sec. 5.2) intimate that diffusion models apprehend counting-based factuality with greater reliability when the underlying visual structure is of low morphological complexity (masked objects,

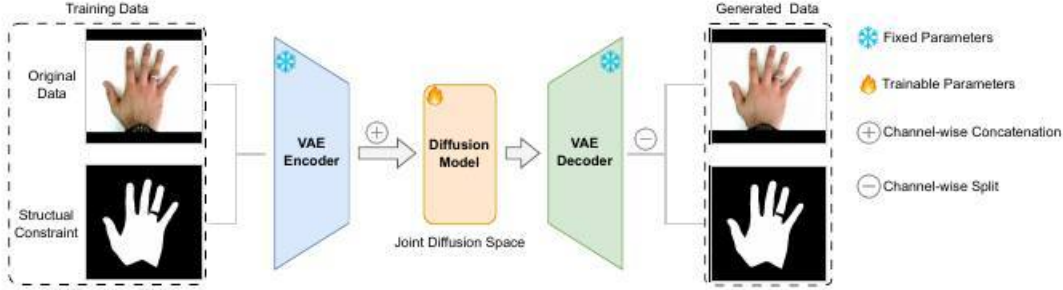


Figure 4: A schematic of the proposed joint-diffusion model (JDM). The model effects a channel-wise concatenation of the latent representations of hand RGB images and their corresponding segmentation masks. This joint learning paradigm fosters anatomically plausible hand poses and promotes correct finger topology throughout generation, with the consequence that both CHR and NCFR are reduced.

Table 3: A comparison of failure rates between the LDM and our proposed JDM on the RealHand dataset across a range of ODE solvers and sampling budgets. Lower values denote superior performance. For JDM, although the model jointly generates hand RGB images and segmentation masks, FID is computed exclusively on the RGB outputs so as to preserve comparability with the baselines.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

for example). This finding, in turn, prompts a consequential question: were diffusion models to denoise general visual representations in concert with explicit, readily modeled structural constraints—object masks encoding rudimentary topology and counts, say—within a shared latent space, might such a joint learning paradigm serve to mitigate counting hallucinations? Models that thus jointly learn both general visual representations and structural constraints over the course of denoising we designate **joint-diffusion models** (JDM).

Concretely, we effect a channel-wise concatenation of the latent representation of the hand mask with that of the corresponding RGB image, as shown in Fig. 4. We then fine-tune the UNet pretrained on CelebA-HQ (Karras et al., 2018) (in accordance with the RealHand configuration set out in Sec. 5.1) upon the RealHand images together with their associated masks, the latter extracted by means of SAM-2 (Ravi et al., 2024). So as to accommodate both modalities, the input and output channels of the denoising UNet are expanded to match the combined latent dimensionality of the image and mask latents (i.e., $3 + 3 = 6$ channels).

As Table 3 shows, JDM consistently attains lower failure rates—encompassing both CHR and NCFR—than the vanilla LDM across a diversity of sampling conditions, thereby evidencing its efficacy in the faithful generation of structurally intricate objects that are prone to counting ambiguity. JDM, on the other hand, incurs a higher FID. This degradation plausibly issues from the augmented denoising burden that attends the joint modeling of two modalities under a small-scale training regime, which constrains the model to forgo a measure of perceptual quality in exchange for improved structural realism. We further observe auspicious reductions in counting hallucinations upon applying JDM to large-scale pretrained diffusion models under carefully controlled fine-tuning settings. Additional details, results, and discussion are consigned to Appendix A.9.

6 CONCLUSION

In this work we take a first step toward the systematic quantification of hallucinations in diffusion models and, in so doing, furnish new insight into the improvement of factual generation within the field. Concentrating upon counting hallucinations, we introduce **CountHalluSet** together with a practical and reproducible methodology for their measurement in practice. Through comprehensive and controlled quantification experiments, we establish that the numerical strategies customarily adopted frequently fail to attenuate counting hallucinations in the generation of human hands. Our correlation

analyses demonstrate, moreover, that the perceptual metrics in widest use—FID and MUSIQ among them—prove incapable of consistently reflecting the severity of counting hallucinations. Experiments on large-scale pretrained text-to-image models further corroborate both the generality and the validity of our findings.

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A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the true reverse process $p_\theta(\mathbf{x}_{T:0})$, the KL divergence between them is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Invoking the Markov property, we may decompose both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$ as follows:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule to decompose the KL divergence, we obtain:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

By the Data Processing Inequality (Cover & Thomas, 2012), marginalization—the discarding of certain random variables—cannot increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality would obtain strictly only in the idealized case in which $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$ —that is, precisely when the diffusion model recovers the underlying data distribution $q(\mathbf{x}_0)$ exactly. This upper bound on the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$) between the true and the model-generated data coincides with the variational bound derived in DDPM (Ho et al., 2020) once the data entropy $H(\mathbf{x}_0)$ is excluded, whence it follows that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between true and model-generated data.

Diffused-prior gap. The first source of error arises from the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ and the prior distribution $p_\theta(\mathbf{x}_T)$ employed for sampling. This mismatch is a direct consequence of the number of diffusion steps T being finite in practice. The discrepancy admits formal quantification through a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Alternatively, other suitable metrics—FID or the Wasserstein distance, for instance—may be enlisted to evaluate this distributional discrepancy. The diffused-prior gap, present from the very outset, induces a deviation that persists throughout the denoising process and culminates in a sustained output error.

Noise model prediction error. The second source of error is traceable to the noise prediction model, whose estimate $\epsilon_\theta(\mathbf{x}_t, t)$ merely approximates the true noise ϵ —an approximation guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ denote this prediction error. By Eq. (5), any discrepancy in ϵ_θ induces a commensurate deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step draws upon the mean furnished by the preceding iteration, this bias accumulates along the denoising trajectory and thereby contributes to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Taking Eq. (7) as our point of departure, we consider by way of illustration a one-step numerical solver that approximates the solution. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory satisfying this ODE under the true score function $h^*(\mathbf{x}(\tau), \tau)$ and originating from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Correspondingly, let $\hat{\mathbf{x}}_k$ denote the numerical trajectory computed by the ODE solver at the discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ denote the estimated model function. For a given numerical trajectory spanning $\hat{\mathbf{x}}_0$ through $\hat{\mathbf{x}}_N$, our objective is to characterize its total accumulated trajectory error. To this end, let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k , and consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Invoking the variation-of-constants formula for the exact solution over a single step (from τ_k down to τ_{k+1}) yields:

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator associated with the linear part $\frac{d\mathbf{z}}{d\tau} = f(\tau)\mathbf{z}$; note that the limits of integration run backward in time.

Numerical path evolution. A one-step numerical solver approximates the solution, and this numerical step departs from the exact evolution governed by $\hat{h}$ that issues from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h})$ denote the exact solution at τ_{k+1} of the ODE driven by $\hat{h}$ and initiated from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method thereby introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

Consequently, we obtain:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12) gives:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. This recurrence relation may be unrolled from $k = 0$ to $N - 1$. Letting $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, with $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$, and applying the recurrence repeatedly, we arrive at:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ then furnishes:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated-initial-error term captures the deterministic bearing that a particular instance of initial error exerts upon the final path error, its statistical provenance residing in the distributional diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As Eq. (2) indicates, the forward process in DDPM (Ho et al., 2020) may be formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

The reverse process, in turn, is defined as follows:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process coincides with that of Eq. (A.16), whereas its deterministic reverse process (with $\eta = 0$) is defined as follows:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share an identical forward (diffusion) process, DDIM does not merely excise the stochastic term from DDPM’s reverse (denoising) process; the divergence arises, rather, because the two methods employ distinct coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples drawn from the three datasets employed in this study. As Table A1 attests, each dataset exhibits a balanced distribution across its respective categories, while Table A2 further demonstrates that the distribution of images over differing object counts is comparable, thereby evidencing the absence of any object-count bias across the datasets. The datasets are listed in the same order in both tables, and the category order in Table A1 is defined as follows:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: The number of objects per category within each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: The number of images stratified by object count(s) within each dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We furnish a selection of representative counting hallucinations corresponding to the three datasets, as shown in Fig. A2. For the ToyShape and SimObject datasets, the counting hallucinations either harbor more than two instances of at least one category or, alternatively, contain no instance whatsoever yet retain a normal background, as shown in Fig. A2(a) and Fig. A2(b). In the RealHand dataset, counting-hallucinated samples are distinguished by incorrect finger counts (four or six fingers, for example), as shown in Fig. A2(c). Of particular note, the last two images of the first row in Fig. A2(c) appear at first glance to exhibit five fingers; one finger, however, is partially incomplete and conspicuously artifacted, so that four valid fingers effectively remain. These samples are, in consequence, classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative examples of YOLO predictions on the RealHand dataset. The YOLO detector localizes fingertips accurately across a diversity of hand poses and orientations, spanning both dorsal and palmar views. It may nonetheless fail to apprehend anatomically implausible configurations—hands endowed with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)), or hands displaying five middle fingers yet lacking a thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). Such counting-valid samples are still detected as five-finger hands and therefore remain, as regards finger counts, indistinguishable from their anatomically valid counterparts.

Figure A3(b) exhibits several counting-hallucinated samples for which YOLO predicts erroneous fingertip counts (3, 4, 6, or 7 fingertips, for instance). Figure A3(c) illustrates, in addition, several rare edge cases (<1%) in which the detector fails to identify fingertips beset by unclear boundaries, cracked fingers, or floating artifacts. Such rare cases may be partially ameliorated by tuning YOLO’s inference hyperparameters (the IoU and confidence thresholds, say); doing so, however, risks compromising detection performance on the preponderance of counting-valid samples. Inasmuch as a consistent detection configuration is applied across all evaluations on the RealHand dataset, the presence of these rare cases in no way vitiates the validity of our overall conclusions, even as their remediation remains a challenging and low-reward undertaking.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 exhibits representative examples of the three classes of generated hand image: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the evaluation procedure for the RealHand dataset first applies the counting-ready indicator so as to separate counting-ready samples (namely, the counting-valid and counting-hallucinated) from images unsuitable for counting, the latter typically betraying severe visual artifacts such as distorted, cracked, partially

visible, or floating fingers. Counting-ready samples are thereafter further classified as counting-valid or counting-hallucinated according to whether the number of detected valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we adopt 10 distinct hand-generation prompts with which to prompt Stable-Diffusion-3.5-Medium. The prompts are enumerated below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person's hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The definition of hallucination presupposes that the model generates patterns never observed during training. This presupposition, however, proves difficult to verify in the case of large-scale pretrained diffusion models, for which the full training distribution is inaccessible. To render the quantification of counting hallucinations tractable in such circumstances, we adopt a conservative criterion, treating any generated hand image that contains six or more fingers as a counting-hallucinated sample. This criterion is well warranted: whereas real images may depict fewer than five visible fingers on account of occlusion, articulation, or viewpoint, it is exceedingly improbable that real data should contain hands with six or more fingers. A finger count of ≥ 6 thus constitutes a reliable and empirically verifiable signal of counting hallucination. Generated examples are provided in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, as against 80k steps for the vanilla LDM. This enlarged training budget is rendered necessary by the circumstance that extending the latent space from three to six channels in JDM introduces a substantial mismatch with the pretrained latent distribution of the LDM. Additional optimization is consequently required to bridge this distributional gap and to permit JDM's modeling capacity to materialize in full.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

It bears emphasis that JDM ought to be regarded as **a novel training paradigm** for the improvement of factual generation in diffusion models, rather than as a superior inference strategy. On the one hand, JDM merely extends the input and output channels of the UNet while leaving the dimensions of all intermediate layers unaltered, and thus **introduces a negligible inference-time overhead relative to the vanilla LDM**. On the other hand, augmenting a pretrained text-to-image model with supplementary mask channels **perturbs the manifold geometry of the pretrained latent space and**

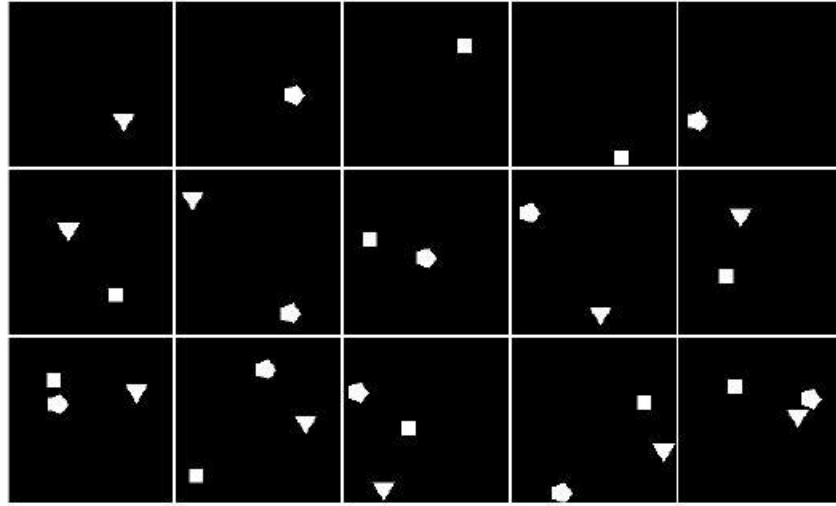
Table A3: A comparison of failure rates and image quality between fine-tuning SD-1.5 and fine-tuning SD-1.5 under the joint-diffusion training paradigm on RealHand. The first row (baseline) reports results obtained from fine-tuning Stable-Diffusion-1.5 on RealHand data. The second row reports results obtained by freezing all intermediate layers of the baseline model and fine-tuning only the input and output layers with extended channel dimensions. Lower values denote superior performance.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

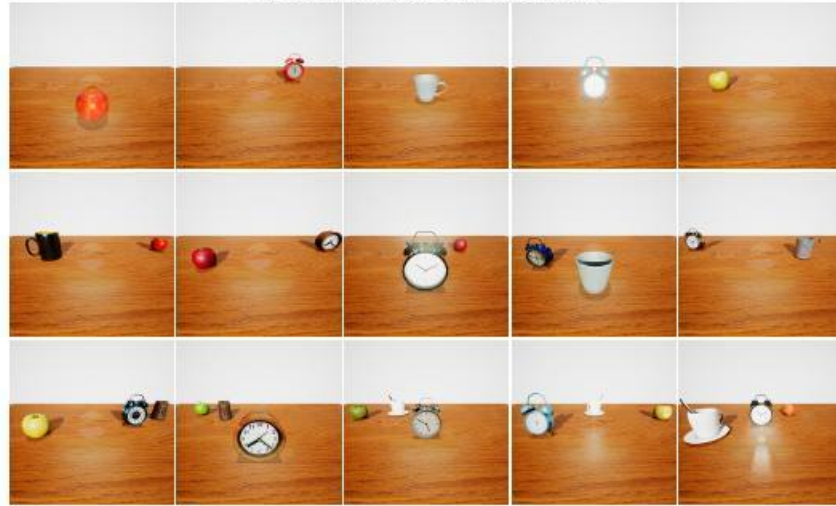
undermines the original denoising distribution, whereby the direct fine-tuning of such a model is rendered far from trivial.

This challenge notwithstanding, we still observe auspicious reductions in counting hallucinations upon applying JDM to large-scale pretrained diffusion models under carefully controlled fine-tuning conditions. In practice, we find that jointly reconciling the extended latent space of JDM with the highly rigid latent geometry of large pretrained models proves challenging given the limited scale of RealHand. To strike a workable balance, we adopt a two-stage fine-tuning strategy: we first fine-tune a Stable-Diffusion-1.5 model on RealHand, and thereafter freeze all intermediate UNet layers, fine-tuning only the input and output layers with their extended channels. As Table A3 shows, even under this severely constrained fine-tuning regime, the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps across both solvers.

Taken as a whole, the evidence adduced in this work indicates that JDM constitutes a promising avenue toward enforcing factual consistency in image generation. We hold, finally, that **the investigation of its behavior and performance under genuine large-scale training remains an important and compelling direction for future research**.



(a) Examples in the ToyShape dataset

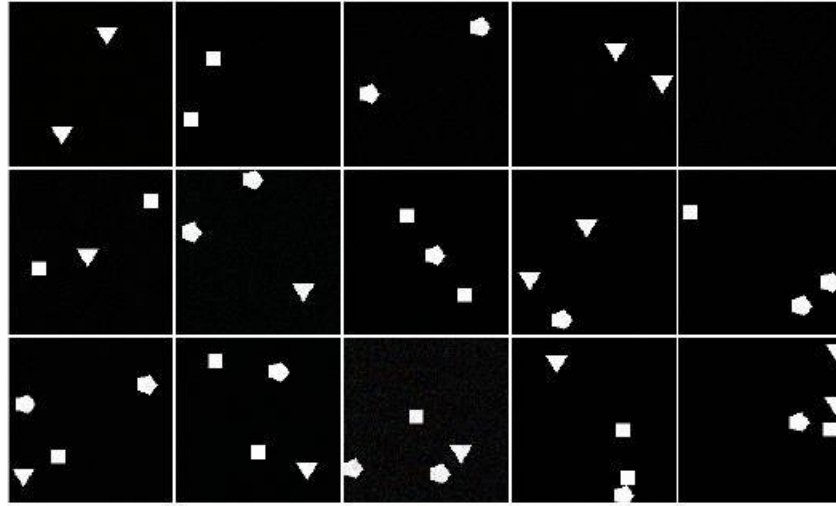


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

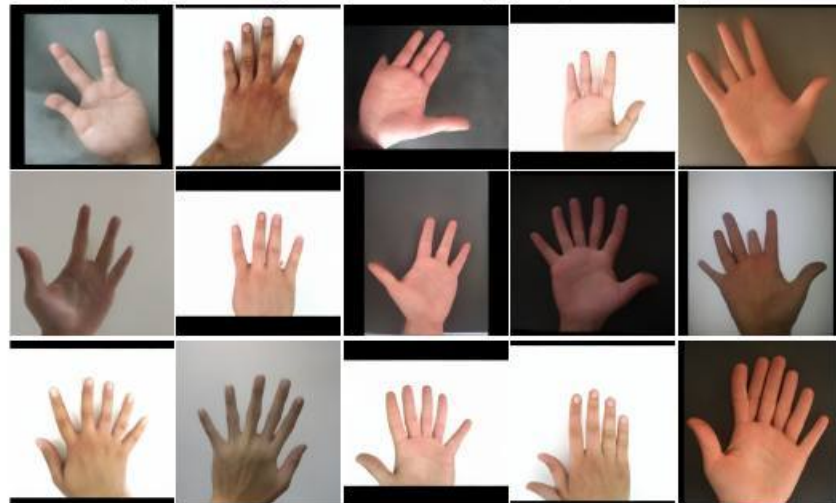
Figure A1: Representative examples of the datasets constituting the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

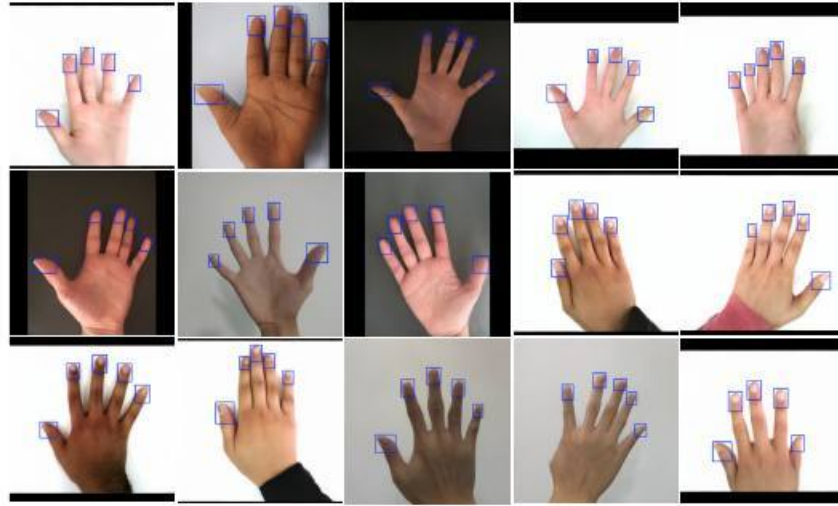


(b) Examples of counting hallucinations corresponding to the SimObject dataset

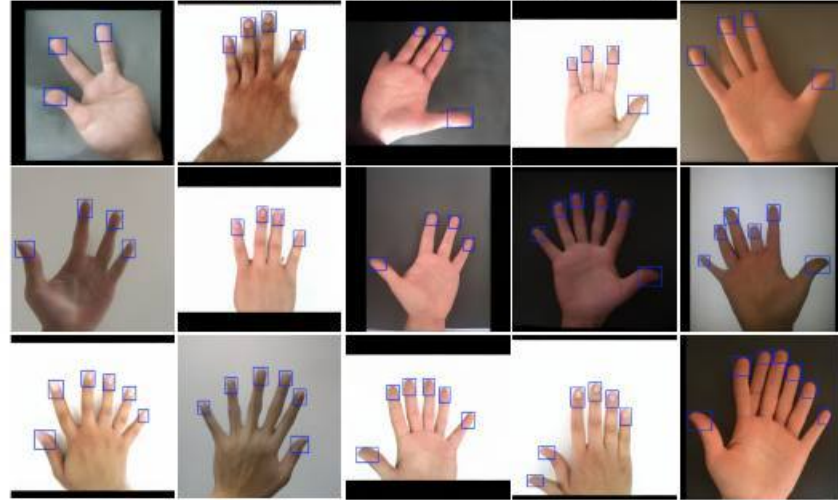


(c) Examples of counting hallucinations corresponding to the RealHand dataset

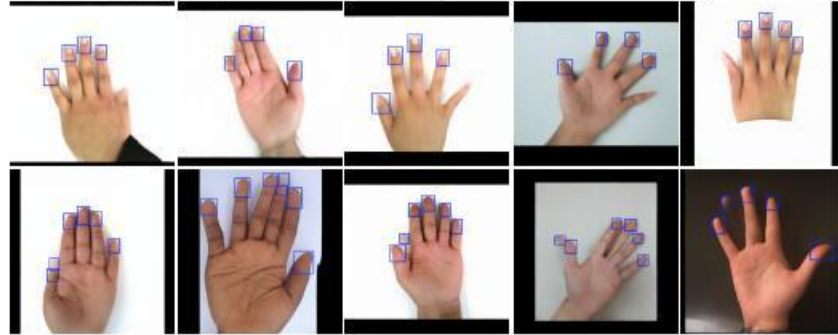
Figure A2: Representative counting hallucinations corresponding to the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers



(b) Counting-hallucinated YOLO predictions with not five fingers



(c) Rare edge cases where YOLO predictions fail

Figure A3: Representative YOLO predictions on generated RealHand images. (a) Counting-valid samples for which YOLO correctly predicts five fingertip bounding boxes (including cases exhibiting two thumbs). (b) Counting-hallucinated samples for which the predicted fingertip count is erroneous (e.g., 3, 4, 6, or 7 detected fingertips). (c) Several rare edge cases that the counting-ready indicator fails to filter out and that YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.

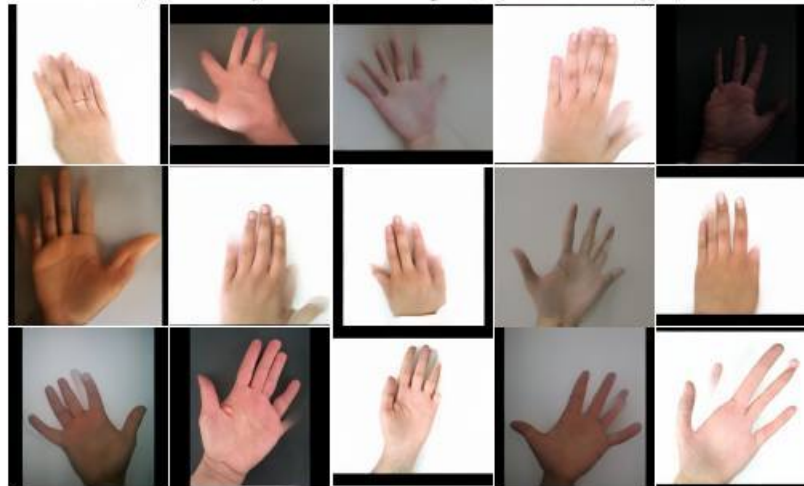
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(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Representative examples of the three classes of generated hand image. (a) Counting-valid: images bearing five clearly discernible fingers. (b) Counting-hallucinated: images bearing an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images rendered unsuitable for counting by severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Representative examples of the three classes of generated hand image produced by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images bearing five clearly discernible fingers. (b) Counting-hallucinated: images bearing more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images rendered unsuitable for counting by severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.